

HOSPITAL MANAGEMENT SYSTEM

OBJECTIVE

To develop a Hospital Management System using C language for managing patient records.

CONCEPTS USED

Structures

Arrays

Loops

Switch Case

CRUD Operations

SOURCE CODE

```
#include <stdio.h>
```

```
struct Patient {
```

```
    int id;
```

```
    char name[50];
```

```
    int age;
```

```
};
```

```
int main() {
```

```
    struct Patient p[100];
```

```
    int n = 0, choice, id, i, found;
```

```
    while(1) {
```

```
        printf("\n===== Hospital Management System =====\n");
```

```
        printf("1. Add Patient\n");
```

```
        printf("2. View Patients\n");
```

```
        printf("3. Search Patient\n");
```

```
printf("4. Update Patient\n");  
printf("5. Delete Patient\n");  
printf("6. Exit\n");
```

```
printf("Enter Choice: ");  
scanf("%d", &choice);
```

```
switch(choice) {
```

```
case 1:
```

```
    printf("Enter Patient ID: ");  
    scanf("%d", &p[n].id);
```

```
    printf("Enter Name: ");  
    scanf(" %[^\\n]", p[n].name);
```

```
    printf("Enter Age: ");  
    scanf("%d", &p[n].age);
```

```
    n++;
```

```
    printf("Patient Added Successfully! \n");  
    break;
```

```
case 2:
```

```
    if(n == 0) {  
        printf("No Patient Records Found!\n");  
    }  
    else {
```

```
printf("\n Patient Records \n");
```

```
for(i = 0; i < n; i++) {
```

```
    printf("\nPatient %d\n", i + 1);
```

```
    printf("ID   : %d\n", p[i].id);
```

```
    printf("Name : %s\n", p[i].name);
```

```
    printf("Age  : %d\n", p[i].age);
```

```
}
```

```
}
```

```
break;
```

case 3:

```
printf("Enter Patient ID to Search: ");
```

```
scanf("%d", &id);
```

```
found = 0;
```

```
for(i = 0; i < n; i++) {
```

```
    if(p[i].id == id) {
```

```
        printf("\nPatient Found\n");
```

```
        printf("ID   : %d\n", p[i].id);
```

```
        printf("Name : %s\n", p[i].name);
```

```
        printf("Age  : %d\n", p[i].age);
```

```
        found = 1;
```

```
        break;
    }
}
```

```
if(found == 0)
    printf("Patient Not Found!\n");
```

```
break;
```

case 4:

```
printf("Enter Patient ID to Update: ");
scanf("%d", &id);
```

```
found = 0;
```

```
for(i = 0; i < n; i++) {
```

```
    if(p[i].id == id) {
```

```
        printf("Enter New Name: ");
        scanf(" %[^\n]", p[i].name);
```

```
        printf("Enter New Age: ");
        scanf("%d", &p[i].age);
```

```
        printf("Patient Updated Successfully!\n");
```

```
        found = 1;
        break;
    }
```

```
}
```

```
if(found == 0)
```

```
    printf("Patient Not Found!\n");
```

```
break;
```

case 5:

```
printf("Enter Patient ID to Delete: ");
```

```
scanf("%d", &id);
```

```
found = 0;
```

```
for(i = 0; i < n; i++) {
```

```
    if(p[i].id == id) {
```

```
        int j;
```

```
        for(j = i; j < n - 1; j++) {
```

```
            p[j] = p[j + 1];
```

```
        }
```

```
        n--;
```

```
        printf("Patient Deleted Successfully!\n");
```

```
        found = 1;
```

```
        break;
```

```
    }
```

```

    }

    if(found == 0)
        printf("Patient Not Found!\n");

    break;

case 6:

    printf("Thank You!\n");
    return 0;

default:

    printf("Invalid Choice!\n");
}
}

return 0;
}

```

OUTPUT

===== Hospital Management System =====

1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 1

Enter Patient ID: 10000001

Enter Name: RAMU

Enter Age: 28

Patient Added Successfully

===== Hospital Management System =====

1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 1

Enter Patient ID: 002

Enter Name: AMMULYA

Enter Age: 34

Patient Added Successfully!

===== Hospital Management System =====

1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 5

Enter Patient ID to Update: 002

Enter New Name: AMMULYAA

Enter New Age: 34

Patient Updated Successfully!

===== Hospital Management System =====

1. Add Patient

2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 5

Enter Patient ID to Delete: 001

Patient Deleted Successfully!

===== Hospital Management System =====

1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 3

Enter Patient ID to Search: 002

Patient Found

ID : 2

Name : AMMULYAA

Age : 34

===== Hospital Management System =====

1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Exit

Enter Choice: 6

Thank You!

CONCLUSION

The Hospital Management System was successfully implemented using C language. Add, View, Search, Update and Delete operations were performed successfully.